

Java OOP – Learning Guide

1. Classes and Objects

A class is a blueprint containing data and methods. An object is an instance of a class.

Example: `class Student { String name; void display() { System.out.println(name); } }`

2. Encapsulation

Encapsulation combines data and methods inside a class and controls access using access modifiers. private fields are commonly accessed through public getter and setter methods.

3. Inheritance

Inheritance allows a child class to acquire accessible members of a parent class. Java uses the `extends` keyword for class inheritance.

Example: `class Dog extends Animal { }`

4. Polymorphism

Polymorphism means one interface or method name can represent different behavior. Method overloading is compile-time polymorphism, while method overriding is commonly associated with runtime polymorphism.

5. Method Overloading

Method overloading means using the same method name with different parameter lists in the same class.

Example: `add(int a, int b)` and `add(int a, int b, int c)`.

6. Method Overriding

Method overriding occurs when a subclass provides its own implementation of an inherited method with a compatible signature.

7. Abstraction

Abstraction hides unnecessary implementation details and exposes essential behavior. Java supports abstraction through abstract classes and interfaces.

8. Important Keywords

`extends` is used for class inheritance. `implements` is used when a class implements an interface. `super` refers to members of the parent class. `this` refers to the current object.